

Quantum Computing in Robotics: A Review of Optimization, Search, and Machine Learning Approaches

Daniel Udekwe^{a,*}

^aDepartment of Civil & Environmental Engineering, FAMU-FSU College of Engineering, Tallahassee, 32310, Florida, USA

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ABSTRACT

Quantum computing has begun to intersect with robotics along several distinct lines: quantum optimization for planning and control, quantum search and sampling for decision making, and quantum machine learning for perception and policy learning. This review organizes the emerging literature through a systematic search of the quantum robotics literature guided by four research questions, first recasting core robotics tasks as computational problems, namely combinatorial optimization, search, sampling, and linear algebra, and then surveying how quantum and quantum-inspired methods have been applied to each. We examine quantum annealing and variational approaches to robotic control, Grover-type search and quantum-inspired sampling for decision making, and quantum machine learning for perception and reinforcement learning, before turning to the hardware, latency, and systems constraints that currently bound real-world deployment, including quantum sensing as a complementary technology. The core quantum primitives underlying these approaches, including the QUBO and Ising formulations, Grover's algorithm, and the QAOA and variational circuit ansätze, are formalized mathematically and illustrated through representative circuit architectures. We conclude by identifying open challenges, including the near-total reliance on simulation rather than physical validation, the absence of standardized benchmarks, the persistent ambiguity between genuinely quantum and merely quantum-inspired methods, and the mismatch between current noisy intermediate-scale quantum hardware and the real-time demands of robotic control loops.

1. Introduction

Robotics has historically drawn its computational tools from classical optimization, probabilistic estimation, and, more recently, deep learning. As robotic systems scale in the number of degrees of freedom they control, the number of agents they coordinate, and the uncertainty of the environments they operate in, many of the underlying problems, including multi robot task allocation, inverse kinematics, simultaneous localization and mapping, and policy learning, retain combinatorial or search structure that is NP hard or otherwise intractable at scale for classical solvers. This has motivated a growing body of work exploring whether quantum computing, still in its noisy intermediate scale quantum (NISQ) era, can offer meaningful computational leverage for robotics.

The appeal is not purely theoretical. Quantum annealers such as D-Wave's Advantage platform are commercially accessible today and have already been applied to robot path planning, task allocation, and inverse kinematics formulated as quadratic unconstrained binary optimization (QUBO) problems [113, 137]. Gate based quantum processors, though far more resource constrained, have been used to demonstrate Grover accelerated search for manipulator configuration optimization [105] and variational quantum circuits for reinforcement learning based navigation [54]. At the same time, a distinct and older thread of quantum inspired classical algorithms, including quantum behaved particle swarm optimization, quantum genetic algorithms, and

quantum inspired reinforcement learning, has used quantum formalism as a source of heuristics for classical robot control and estimation problems without requiring quantum hardware at all [37, 68, 175].

This diversity of approaches, hardware assumptions, and even definitions of the word quantum has produced a fragmented literature. Several recent surveys have begun to map parts of this space: Mannone and colleagues review quantum computing approaches to swarm robotics specifically [93], Fazilat and Zioui survey quantum based control strategies [45], and Cho and Kim offer a compact introduction to the field as a whole, situating quantum sensing, quantum reinforcement learning, and kinematic optimization within a single classical robotic stack augmented by quantum resources [28]. Petschnigg and colleagues provided an early position paper outlining where quantum computing might plausibly intersect with robotic perception, planning, and control [121], a framing echoed a decade later by a broader review of emerging trends in quantum robotics [164] and by Morasso, the latter situating quantum computing within a longer cybernetic tradition of modeling purposive action and cognition in autonomous agents [101].

Building on these efforts, this review organizes the literature not by algorithm family alone, but by first asking what computational role each robotics problem plays, namely optimization, search, sampling, or learning, and then examining how quantum and quantum inspired methods have been proposed for that role. This framing is intended to clarify which advances in the quantum computing literature are actually transferable to a given robotics problem, and which are not.

*Corresponding author

 dau24@fsu.edu (D. Udekwe)

ORCID(s):

Table 1

Comparison of this review with the closest existing surveys of quantum computing in robotics, scored against the four research questions posed above. RQ1: quantum optimization for planning, allocation, and kinematics. RQ2: quantum search and quantum inspired sampling for decision making and state estimation. RQ3: quantum machine learning for perception and control. RQ4: hardware, latency, and quantum sensing constraints.

Survey	RQ1	RQ2	RQ3	RQ4	Systematic Screening	Key Differentiator from This Review
[121]	✓		✓	✓		Early position paper; does not organize by computational primitive or provide summary tables of individual studies
[93]	✓	✓				Narrower scope limited to swarm coordination; does not cover manipulator kinematics, SLAM, or hardware constraints
[45]	✓		✓			Brief format; summarizes control strategies without distinguishing quantum executed from quantum inspired methods
[164]	✓	✓				Broad descriptive coverage without a computational-primitive taxonomy or quantitative citation trend analysis
[28]	✓		✓	✓		Compact introduction; does not include summary tables, a trend figure, or a systematic search protocol
This review	✓	✓	✓	✓	Planned (PRISMA diagram provided)	Explicit primitive-based taxonomy, per-section summary tables, a citation-year trend figure, a mathematical formulation of the core quantum primitives, and consistent distinction between quantum executed and quantum inspired methods throughout

1.1. Research Questions

This review is guided by four research questions, each aligned with one of the computational primitives introduced in Section 2 and elaborated in the sections that follow.

1. To what extent have quantum optimization methods, particularly quantum annealing and the quantum approximate optimization algorithm, been applied to robotic planning, task allocation, and kinematics problems, and what quantum hardware do these applications assume?
2. How have quantum search algorithms and quantum inspired sampling methods been used to address robotic decision making and state estimation problems such as manipulator configuration search, localization, and simultaneous localization and mapping?
3. What role does quantum machine learning, particularly quantum reinforcement learning, play in robotic perception and control policy learning, and how does its reported performance compare to classical baselines?
4. What hardware, latency, and systems level constraints, including the complementary role of quantum sensing, currently limit the deployment of quantum computing methods in real world robotic systems?

Table 1 situates this review against the closest existing surveys of quantum computing in robotics by scoring each survey's coverage against these same four research questions, rather than merely describing each survey's stated scope in prose. A checkmark indicates that the survey substantively addresses the research question across multiple

studies; an empty cell indicates the question is not addressed or is only mentioned in passing.

1.2. Search Strategy and Study Selection

Sources for this review were identified through a systematic search of five databases, namely Scopus, IEEE Xplore, Web of Science, PubMed, and arXiv, using Boolean combinations of quantum computing terms (including quantum annealing, quantum approximate optimization algorithm, Grover, quadratic unconstrained binary optimization, quantum machine learning, quantum reinforcement learning, variational quantum circuit, and quantum sensing) and robotics terms (including robot, robotic, autonomous system, multi robot, swarm robot, manipulator, unmanned aerial vehicle, simultaneous localization and mapping, motion planning, path planning, robot control, and robot navigation). Records were exported to Rayyan for deduplication and title and abstract screening. Figure 1 summarizes this process following the PRISMA reporting framework.

1.3. Manuscript Structure

The remainder of this paper is organized as follows. Section 2 recasts core robotics tasks as computational problems, establishing the primitives that motivate the four research questions above and the sections that follow. Section 3 addresses the first research question, surveying quantum optimization approaches to robotic control, primarily quantum annealing and QAOA based methods applied to planning, task allocation, and kinematics. Section 4 addresses the second research question, examining quantum search and sampling methods for robotic decision making, centered on Grover type algorithms and quantum inspired sampling for

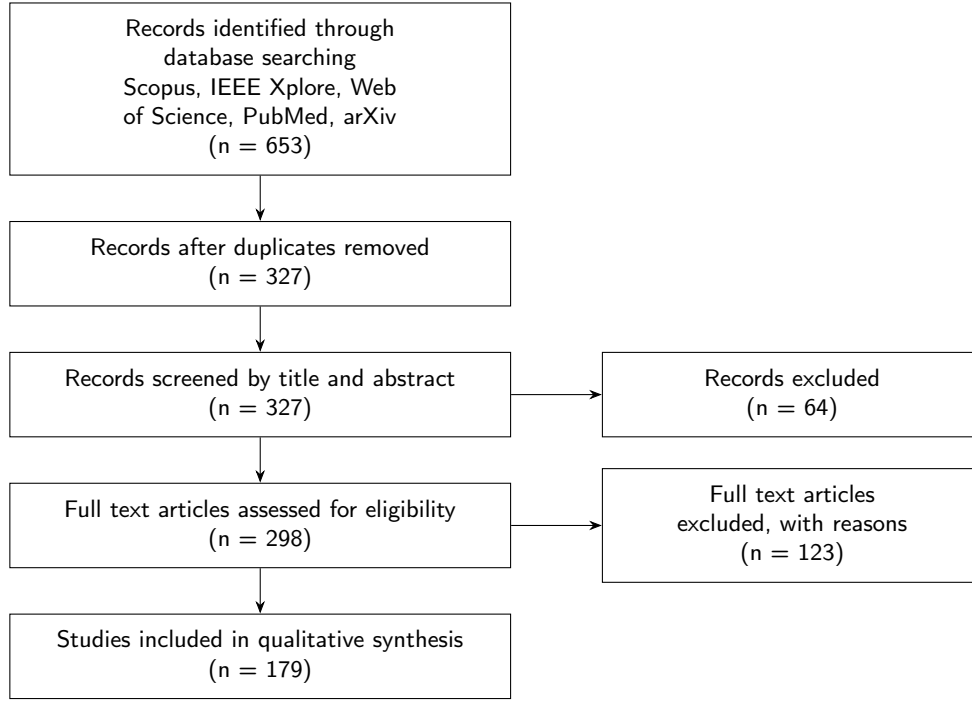


Figure 1: PRISMA flow diagram summarizing the identification, screening, eligibility, and inclusion process used to select the sources synthesized in this review.

estimation problems such as SLAM. Section 5 addresses the third research question, reviewing quantum machine learning in robotics, covering variational quantum circuits for perception and quantum reinforcement learning for control policies. Section 6 addresses the fourth research question, discussing the hardware, latency, and systems constraints, including quantum sensing as a complementary but distinct technology, that bound near term deployment. Section 7 identifies open challenges and future directions, and Section 8 concludes. Fig. 3 summarizes this organization.

Fig. 2 further characterizes the literature synthesized in this review, showing how the sources cited in each technical section are distributed across publication years. The counts are drawn directly from the reference list of this review rather than from an independent bibliometric search, and therefore describe the composition of this survey's evidence base rather than the full population of quantum robotics publications; nonetheless, the pattern is informative, since a large majority of the sources drawn on for optimization and hardware or sensing topics were published in 2024 or later, while the quantum machine learning literature this review draws on has a longer and steadier history dating back to the mid 2000s.

2. Robotic Systems as Computational Problems

Before surveying specific quantum algorithms, it is useful to separate the question of what computational problem a robotics task poses from the question of which quantum

method might address it. Robotics problems that have attracted quantum computing attention cluster into four computational primitives, namely combinatorial optimization, search, sampling, and linear algebra, and framing the literature this way clarifies why certain quantum techniques transfer naturally to certain robotics problems while others, despite superficial appeal, do not.

2.1. Mathematical Preliminaries

To make the technique specific sections that follow self contained, this subsection introduces the small set of mathematical objects that recur throughout the quantum robotics literature: the qubit, the QUBO and Ising formulations that underlie quantum annealing, the query complexity result underlying Grover search, and the variational ansatz underlying QAOA and quantum machine learning.

A single qubit generalizes the classical bit by allowing a linear superposition of the two computational basis states,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1)$$

where α and β are complex probability amplitudes. An n -qubit register is represented in a 2^n -dimensional complex vector space, so that n qubits can in principle represent a superposition over 2^n classical configurations simultaneously; this exponential state space is the source of the representational compactness exploited when, for example, a robot posture with many joints is encoded using a number of qubits that scales with the number of links rather than with the discretized size of the configuration space [113].

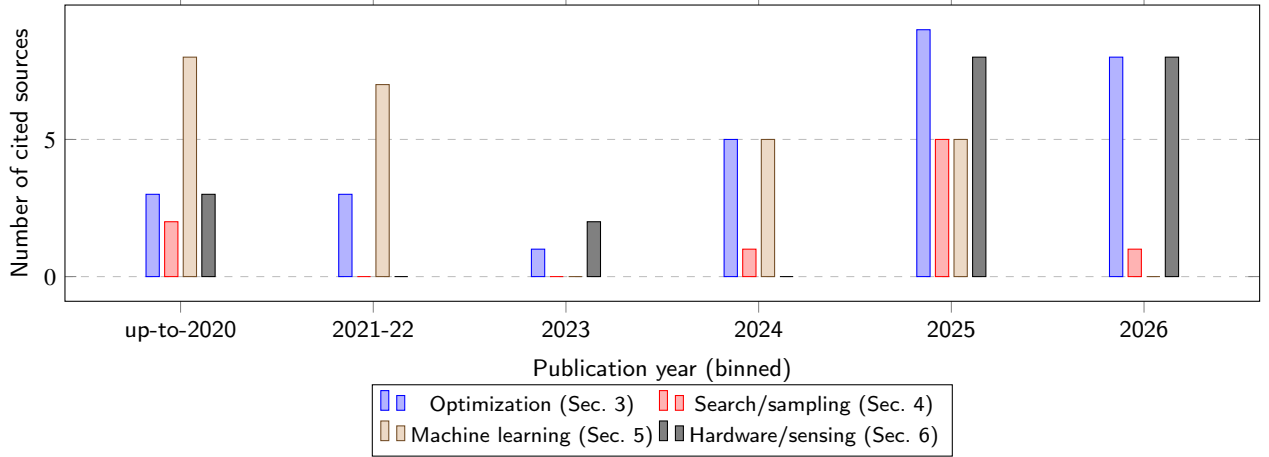


Figure 2: Distribution by publication year of the sources cited in each technical section of this review (Sections 3 through 6). Counts reflect distinct citations in this review's own reference list, not an independent bibliometric census of the field.

Quantum annealing and QAOA both operate on combinatorial optimization problems expressed in quadratic unconstrained binary optimization form,

$$\min_{x \in \{0,1\}^n} x^T Q x = \min_{x \in \{0,1\}^n} \sum_{i=1}^n \sum_{j=1}^n Q_{ij} x_i x_j, \quad (2)$$

where x is a vector of binary decision variables and Q is a matrix encoding both the objective function and any hard constraints as quadratic penalty terms. Equation (2) is the recurring target formulation for the path planning, task allocation, and kinematics problems surveyed in Section 3, since it is the native input format accepted by quantum annealing hardware. An equivalent physical formulation, used directly by annealer hardware such as D-Wave's Advantage platform, recasts the same problem as minimizing an Ising model Hamiltonian over spin variables $s_i \in \{-1, +1\}$,

$$H(s) = \sum_{i < j} J_{ij} s_i s_j + \sum_i h_i s_i, \quad (3)$$

where J_{ij} specifies pairwise coupling strengths and h_i specifies local bias fields; the binary QUBO variables in Equation (2) and the spin variables in Equation (3) are related by the linear substitution $s_i = 2x_i - 1$, and quantum annealing physically evolves a system initialized in a superposition of all spin configurations toward the low energy state of Equation (3) through adiabatic evolution.

Grover's algorithm, underlying the search based approaches surveyed in Section 4, provides a quadratic reduction in query complexity for unstructured search over N candidate configurations,

$$O(\sqrt{N}) \text{ quantum queries} \quad \text{versus} \quad O(N) \text{ classical queries}, \quad (4)$$

achieved by repeatedly applying an oracle that marks target states together with a diffusion operator that amplifies the marked amplitudes, a construction referred to as amplitude

amplification. Concretely, if a fraction M/N of the N candidate configurations are valid targets, for example valid joint configurations reaching a manipulator's goal pose, then after k applications of the oracle-diffuser pair the probability of measuring a target state is

$$P(k) = \sin^2((2k+1)\theta), \quad \theta = \arcsin \sqrt{M/N}, \quad (5)$$

which is maximized, and the search terminates, after approximately $k^* \approx \frac{\pi}{4} \sqrt{N/M}$ iterations. This is the source of the reported speedups for Grover assisted manipulator kinematic optimization and localization discussed in Section 4, and it explains why those results are reported as query complexity reductions rather than as constant factor speedups: the quadratic gain in Equation (4) only manifests at problem sizes N large enough for the square root reduction in Equation (5) to outweigh the overhead of constructing and running the quantum circuit.

Finally, QAOA and the variational quantum circuits used for quantum machine learning in Section 5 share a common ansatz structure in which a parameterized quantum circuit alternates between a problem dependent cost unitary and a mixing unitary,

$$|\gamma, \beta\rangle = U_B(\beta_p) U_C(\gamma_p) \cdots U_B(\beta_1) U_C(\gamma_1) |+\rangle^{\otimes n}, \quad (6)$$

where $U_C(\gamma) = e^{-i\gamma H_C}$ encodes the QUBO or Ising objective from Equations (2)-(3) as a cost Hamiltonian H_C , $U_B(\beta) = e^{-i\beta H_B}$ is a mixing Hamiltonian that drives exploration across candidate solutions, p is the number of alternating layers, and $|+\rangle^{\otimes n}$ is the uniform superposition over all n -qubit basis states. The variational parameters γ and β are tuned by a classical optimizer in a hybrid quantum-classical loop, the same loop structure that underlies the variational quantum reinforcement learning policies and quantum machine learning classifiers surveyed in Section 5, where the cost Hamiltonian is replaced by a task specific loss function and the circuit parameters play the role of trainable neural network weights. In the quantum reinforcement learning

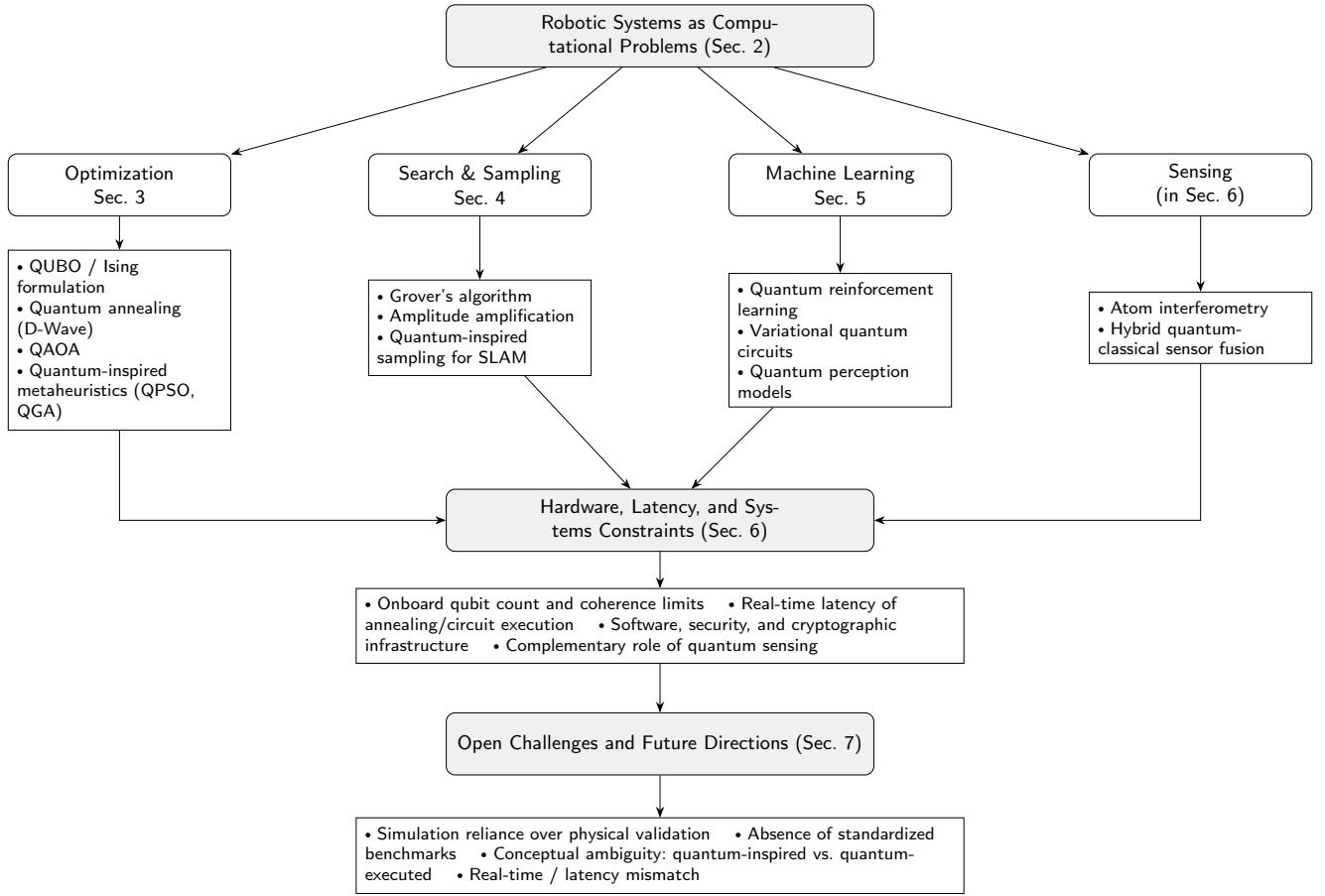


Figure 3: Overview of the organizing framework used in this review. Robotics tasks are first recast as computational primitives (Section 2), which motivate three parallel technique based sections (optimization, search and sampling, and machine learning), with quantum sensing addressed as a complementary hardware layer within Section 6 alongside latency and systems constraints. Sub-item lists indicate the principal techniques or issues addressed within each branch. Findings across all technique sections converge on a common set of open challenges (Section 7).

setting specifically, a parameterized quantum circuit $U(\theta)$ acting on a data encoded input state $|\phi(s)\rangle$ for robot state s produces an expectation value read out from a chosen observable O , most often a Pauli-Z operator on one or more qubits,

$$Q_\theta(s, a) \text{ or } \pi_\theta(a | s) \propto \langle \phi(s) | U^\dagger(\theta) O U(\theta) | \phi(s) \rangle, \quad (7)$$

which plays the role of the Q-value or action probability that a classical neural network would otherwise produce, with θ updated by gradient based classical optimization exactly as the QAOA parameters γ, β are updated in Equation (6).

Fig. 4 illustrates the internal gate level structure of the three circuit families defined by Equations (6)-(7), complementing the higher level algorithmic descriptions in Equations (2)-(4).

2.2. Four Computational Primitives

A large share of robotics problems reduce to choosing a discrete configuration, such as a path, an assignment, or a joint angle discretization, that minimizes some cost function subject to constraints. Multi robot task allocation, coverage path planning, robot assembly line balancing, and

even inverse kinematics under discretized joint angles can all be formulated as QUBO or Ising model problems [49, 143], and the quantum approximate optimization algorithm underlying much of the gate based variant of this literature has itself been the subject of dedicated methodological reviews and introductory treatments [17, 51], including proposals to remove the classical parameter optimization loop entirely using a neural network trained to predict good QAOA parameters directly [9]. A related quantum inspired formulation has addressed multimodal container and logistics planning as a QUBO problem, a non-robotic but structurally analogous combinatorial routing problem [122]. This formulation matters because QUBO is the native problem class for quantum annealers, so once a robotics problem is expressed as a QUBO it can in principle be submitted to annealing hardware without further translation. This explains why quantum annealing dominates the quantum optimization literature in robotics far more than gate based approaches do, since the mapping from robotics problem to quantum native format is comparatively direct, even though embedding a large QUBO onto physical annealer topology introduces its own overhead [70, 137].

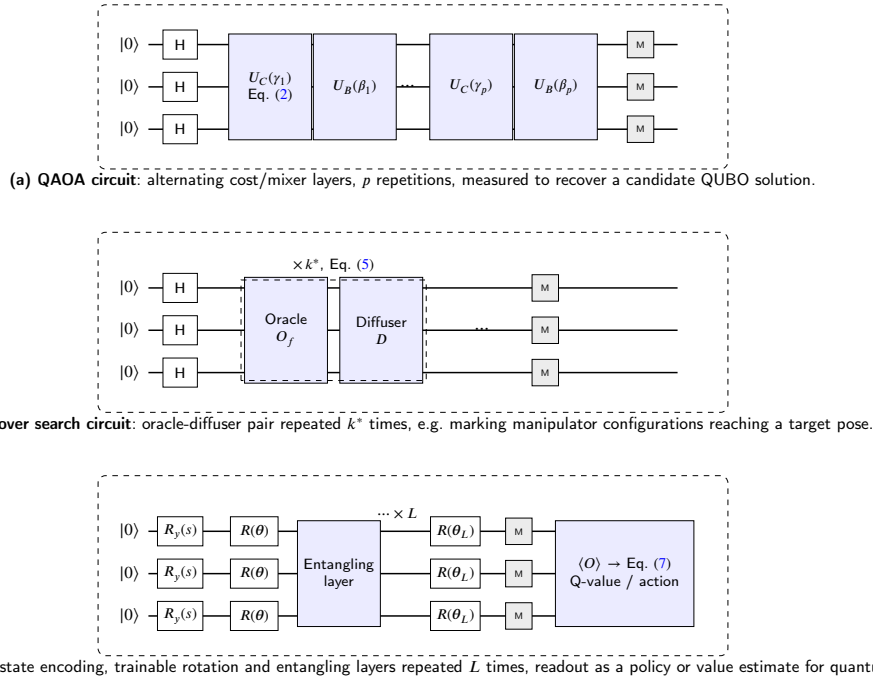


Figure 4: Gate level structure of the three quantum circuit families most frequently used in the robotics literature surveyed in this review: (a) the QAOA circuit underlying quantum optimization approaches in Section 3, (b) the Grover search circuit underlying quantum search approaches in Section 4, and (c) the variational quantum circuit underlying quantum reinforcement learning and quantum machine learning approaches in Section 5. All three follow the parameterized, hybrid quantum-classical structure summarized in Equations (6)-(7).

A second class of robotics problems is not about optimizing a continuous or combinatorial cost function but about finding a specific configuration within an unstructured space, for instance identifying a manipulator joint configuration that achieves a target end effector pose, or localizing a robot within a discretized map. These are search problems in the formal sense: given an oracle that can verify whether a candidate solution is correct, the task is to find that solution with as few oracle queries as possible. This is precisely the structure Grover’s algorithm was designed for [52], complementing the quantum approximate optimization algorithm’s own foundational formulation [43] as one of the two algorithmic primitives most frequently adapted for robotics, and it explains the appearance of Grover based methods for manipulator kinematic optimization [105, 138] and robot localization [10] in the literature, since these are framed as search problems with a quadratic query complexity speedup available in principle, rather than as optimization problems in the QUBO sense.

A third class of problems is inherently probabilistic, such as estimating a robot’s pose and map jointly under sensor and motion uncertainty in SLAM, or maintaining a belief distribution over possible states in a particle filter. These problems require efficient sampling from complex, high dimensional distributions, and it is here that the literature’s use of the word quantum becomes least literal. Much of the published work on so called quantum SLAM does not use quantum hardware at all, but instead borrows the mathematical formalism of quantum mechanics, in the form of

quantum behaved particle swarm optimization, as a classical heuristic for improving the proposal distribution used in particle filter resampling, addressing the particle degeneracy problem that afflicts classical Rao-Blackwellized particle filters [120, 175]. This distinction matters for a review paper, because quantum inspired sampling heuristics are a mature, classically executable line of work, whereas genuinely quantum sampling methods for robotic state estimation remain comparatively unexplored.

A fourth primitive underlies both classical estimation, such as Kalman filtering and forward and inverse kinematics via matrix operations, and machine learning, such as training neural networks and evaluating policies. Quantum linear algebra subroutines and quantum machine learning circuits target this primitive directly, motivating the use of parameterized quantum circuits to approximate forward kinematics as a component of a larger optimization pipeline [105], or to encode robot postures for both forward and inverse kinematics computation on real quantum hardware with as few as one qubit per manipulator link [113]. This same primitive underlies quantum machine learning approaches to reinforcement learning and perception, discussed in Section 5.

Fig. 5 summarizes this four primitive taxonomy in radial form, illustrating for each primitive both the representative robotics tasks it covers and the representative quantum or quantum inspired technique most commonly associated with it in the literature surveyed in Sections 3 through 5.

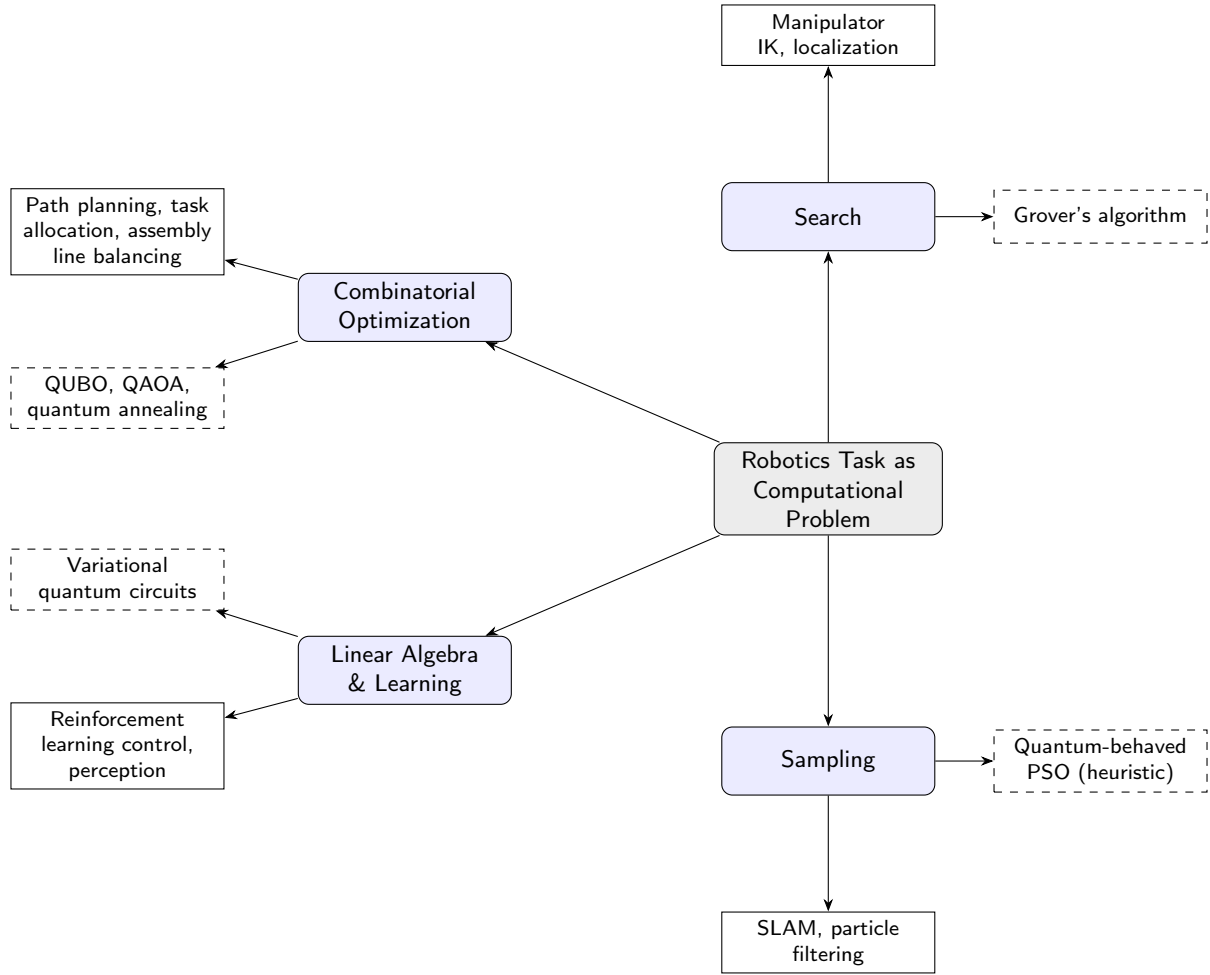


Figure 5: Taxonomy of the four computational primitives underlying quantum approaches to robotics, with representative robotics tasks (solid boxes) and representative quantum or quantum inspired techniques (dashed boxes) associated with each primitive.

2.3. Implications for Hardware and Algorithm Selection

This taxonomy also explains a pattern visible throughout the literature: the choice between quantum annealing and gate based quantum computing in a given paper is rarely arbitrary but tracks the underlying computational primitive. Optimization flavored robotics problems gravitate toward annealing hardware, search and learning flavored problems gravitate toward gate based circuits and variational methods, and sampling flavored problems are, at present, addressed almost entirely by quantum inspired classical heuristics rather than genuine quantum computation. A recurring theme, elaborated in Section 6, is that near term hardware, whether annealer or gate based, imposes size and latency constraints that are frequently the binding limitation on real world deployment, independent of which computational primitive is being targeted [18]. Frameworks such as QUARK have begun to formalize benchmarking practice across these different problem hardware pairings, providing a basis for the kind of standardized comparison this field currently lacks [49].

2.4. A Deployment Framework for Quantum-Enabled Robotics

The individual studies surveyed in Sections 3 through 6 each target one computational primitive in isolation, but it is useful to also see how these pieces would compose into a complete quantum-enabled robotic system. Fig. 6 presents such a composite deployment framework, synthesized from the recurring architectural pattern across the studies surveyed in this review rather than drawn from any single paper. Sensor data from the robot and its environment, including any quantum sensors discussed in Section 6, is first collected and classically preprocessed, since no quantum device in the literature surveyed here accepts raw sensor streams directly. Task relevant features are then encoded into a quantum representation, most often a QUBO matrix for optimization tasks per Equation (2), an oracle circuit for search tasks per Equation (4), or a parameterized feature map for learning tasks per Equation (6). This encoded problem is submitted to whichever quantum resource the study assumes, typically a cloud accessed annealer or gate based processor rather than onboard hardware, and the quantum result, whether an optimized configuration, a search result, or an updated

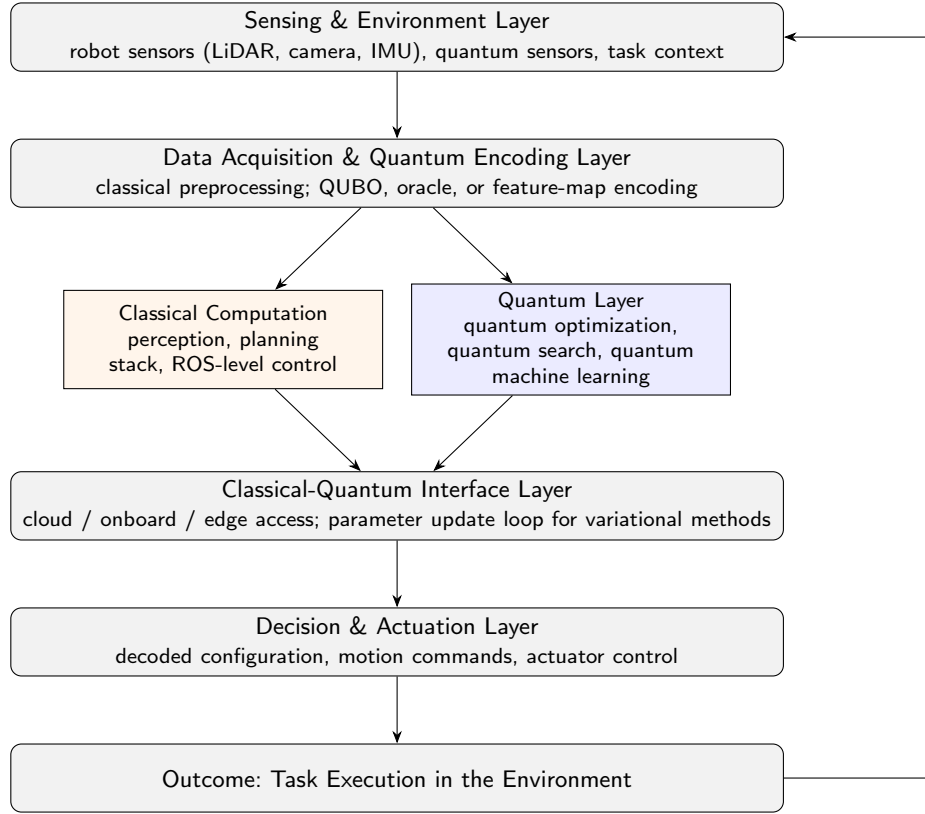


Figure 6: A composite deployment framework for quantum-enabled robotics, synthesized from the recurring architectural pattern across the studies surveyed in this review. The feedback path on the right closes the sense-compute-act loop and, for variational methods, also carries the parameter updates described around Equation (6).

policy, is decoded back into classical form and passed to the robot’s classical control and actuation stack. A classical-quantum interface layer mediates this round trip and, for the variational methods in Equation (6), also closes an outer training loop that updates the circuit parameters based on task performance. The resulting robot action changes the environment and the sensor stream, closing the overall loop. This composite view makes explicit a point argued throughout this review: in essentially every study surveyed here, the quantum computation is one bounded stage within an otherwise classical pipeline, not a replacement for that pipeline.

3. Quantum Optimization for Robotic Control

Quantum annealing is by a wide margin the most mature line of quantum optimization applied to robotics, largely because annealing hardware, most notably D-Wave’s Advantage platform with several thousand qubits, natively accepts problems already expressed as QUBO or Ising models. The generic recipe appearing across dozens of papers is consistent: a robotics planning or control problem is discretized into binary decision variables, a quadratic penalty based cost function is constructed to encode both the objective and any hard constraints, and the resulting QUBO is submitted either directly to annealing hardware or to a hybrid classical

quantum solver that partitions large problems into annealer sized sub-QUBOs [49, 143].

To make this recipe concrete, consider a representative instantiation for multi robot task allocation, the setting addressed in the QAOA based multi drone assignment work of Bang [13] and in related multi robot scheduling work [129]. Let $x_{ij} \in \{0, 1\}$ indicate that robot i is assigned to task j , with cost c_{ij} for that assignment. The assignment objective and the one-task-per-robot constraint are jointly encoded as

$$H_{\text{assign}} = \underbrace{\sum_{i,j} c_{ij} x_{ij}}_{\text{assignment cost}} + \lambda \underbrace{\sum_i \left(1 - \sum_j x_{ij}\right)^2}_{\text{one-task-per-robot penalty}}, \quad (8)$$

where the penalty coefficient λ must be tuned large enough to make constraint violation always more costly than any feasible assignment, but not so large that it distorts the annealer’s energy landscape and degrades solution quality; this tuning step is a recurring, largely manual source of difficulty across the QUBO formulations surveyed in this section. Once cast in the form of Equation (8), the problem matches the general QUBO template of Equation (2) and can be submitted to annealing hardware, which physically implements the time dependent Hamiltonian

$$H(t) = \left(1 - \frac{t}{T}\right) H_0 + \frac{t}{T} H_{\text{Ising}}, \quad t \in [0, T], \quad (9)$$

where H_0 is an easily prepared initial Hamiltonian, typically a uniform transverse field, H_{Ising} is the problem Hamiltonian of Equation (3) encoding Equation (8), and T is the total annealing time. The adiabatic theorem guarantees that a system initialized in the ground state of H_0 remains in the instantaneous ground state of $H(t)$ as t evolves from 0 to T , provided T is large relative to the inverse of the minimum spectral gap encountered along the schedule; in practice this minimum gap, and hence the annealing time actually required for a reliable solution, is unknown in advance for a general QUBO instance, which is one of the reasons that time-to-solution rather than a fixed annealing time is reported in the empirical studies summarized in Table 2. Algorithm 1 summarizes the resulting end-to-end pipeline shared across the QUBO based studies in this section.

Algorithm 1 Generic QUBO-Based Pipeline for Robotic Optimization

- 1: **Input:** robotics planning or control problem P (path planning, task allocation, or kinematics)
 - 2: Discretize P into binary decision variables $x \in \{0, 1\}^n$
 - 3: Formulate objective terms and constraint penalty terms as in Equation (8)
 - 4: Assemble the QUBO matrix Q such that $x^T Q x$ reproduces the objective plus penalties (Equation (2))
 - 5: Tune penalty coefficients λ so that infeasible solutions are always penalized above feasible ones
 - 6: **if** n exceeds the annealer's native qubit connectivity **then**
 - 7: Partition Q into annealer-sized sub-QUBOs and solve iteratively (hybrid solver path)
 - 8: **else**
 - 9: Embed Q directly onto the annealer's physical qubit topology
 - 10: **end if**
 - 11: Execute quantum annealing according to the schedule in Equation (9)
 - 12: Read out the lowest-energy bitstring x^* across the requested number of anneal cycles
 - 13: Decode x^* back into the corresponding robot path, assignment, or configuration
 - 14: **Output:** decoded robotic solution and reported time-to-solution
-

Table 2 summarizes representative studies discussed in this section, organized by robotics task and quantum method.

3.1. Path, Motion, and Target Tracking

Path and motion planning has attracted the largest share of this work. Multi robot path finding has been formulated as a QUBO and benchmarked against classical A* and Dijkstra baselines on grid environments [156], while coverage path planning for multiple robots has been evaluated using both QAOA and simulated annealing against a classical depth first search baseline [131]. In the aerial domain, UAV path planning with obstacle avoidance has been formulated as a QAOA based QUBO and shown to produce shorter paths

than classical rapidly exploring random trees in simulated scenarios [63], and a related line of work uses coherent Ising machines on optical quantum hardware to solve UAV path planning QUBOs, reporting improvements over tabu search and simulated annealing as the number of target points grows [125]. Quantum annealing has also been applied directly to drone route planning framed as a generalized traveling salesman problem [59], to path integral Monte Carlo based clustering and routing of UAV networks with reported reductions in energy consumption relative to simulated annealing [170], and to joint UAV trajectory and beamforming optimization for wireless sensing tasks [42]. A hybrid particle swarm and QUBO framework has similarly been proposed for cooperative defensive screening by multi UAV systems, decomposing the global assignment problem into local trajectory planning and global task allocation stages [117].

Target tracking and real time motion planning represent a related application, where quantum annealing has been used to solve constrained quadratic models for three dimensional target chaser tracking scenarios on D-Wave hardware, framed explicitly as bridging the gap between quantum optimization and time sensitive robotic motion planning [14]. Quantum annealing has also been applied to the dynamics of robot inspectors traversing electrical transmission line conductors, transforming the differential equations governing robot motion into Ising model Hamiltonians suitable for annealing hardware [136], and to automatic path planning for robotic quality inspection tasks derived from computer aided design models in industrial settings, where hybrid quantum classical solvers were compared against classical solvers such as Gurobi and Google OR-Tools on a three dimensional variant of the traveling salesman problem [111]. A related hybrid quantum architecture has been proposed for automated path formation in quantum enabled autonomous mobile robots using a quantum implementation of ant colony optimization that avoids the manual classical to qubit conversion step that limits automation in prior quantum ant colony optimization proposals [139]. Energy efficiency has also been targeted directly, with quantum computing proposed to accelerate the combinatorial search for energy efficient manipulator trajectories under task deadline constraints, reporting meaningful reductions in both torque consumption and computation time [112].

Beyond the D-Wave centric literature discussed above, a substantial and largely independent body of work has applied quantum inspired optimization to surface, aerial, and underwater vehicle path planning, with roots extending back over a decade before the recent wave of quantum annealing hardware became accessible. Quantum ant colony algorithms have been used for global path planning of unmanned surface vehicles [160], quantum inspired monarch butterfly optimization has been applied to unmanned combat aerial vehicle path planning [167], and a quantum inspired firefly algorithm integrated with cuckoo search has been proposed as a general purpose path planning optimizer [82]. Real underwater vehicle path planning has also been explored using quantum potential speed up of combinatorial

Table 2

Representative quantum optimization studies for robotic control. Method type indicates whether the study executes on genuine quantum hardware (annealer or gate based) or uses a quantum inspired classical heuristic.

Task	Year	Study	Method Type	Key Idea
Path planning	2026	[156]	Quantum annealing (QUBO)	Multi robot path finding benchmarked against classical A* and Dijkstra
Path planning	2024	[131]	QAOA / simulated annealing	Coverage path planning versus classical DFS baseline
UAV path planning	2025	[63]	QAOA based QUBO	Obstacle avoidance path planning, shorter paths than classical RRT
UAV path planning	2025	[125]	Coherent Ising machine	Optical quantum hardware for path planning QUBOs
UAV routing	2024	[59]	Quantum annealing	Drone route planning as generalized TSP
Task allocation	2026	[13]	QAOA (W-state, XY-mixer)	Natural language guided multi drone assignment
Task scheduling	2026	[129]	QAOA / VQE	Multi robot scheduling under real time constraints
Assembly line balancing	2024	[159]	Quantum annealing (QBSolv)	Robotic assembly line optimization
Inverse kinematics	2025	[137]	Quantum annealing (QUBO)	Planar IK via binary joint angle discretization
Robot design	2026	[70]	Quantum annealing (QUBO)	Kinematic structure selection for a robotic hand
Posture optimization	2025	[113]	Gate based quantum circuits	Forward/inverse kinematics with qubit count scaling with links
Target tracking	2025	[14]	Quantum annealing (CQM)	3D target chaser tracking on D-Wave hardware
Path planning (heuristic)	2010	[149]	Quantum inspired genetic algorithm	Mobile robot path planning, no quantum hardware
Coalition formation	2009	[172]	Quantum inspired ant colony optimization	Robot coalition formation, no quantum hardware

optimization solvers for autonomous underwater vehicles [100], while cooperative path planning for UAV formations has been addressed with quantum annealing based methods [173] and evolutionary sink node path planning guided by the Hamiltonian formulation used in quantum annealing algorithms [61]. Parallel quantum implementations of the classical rapidly exploring random tree algorithm have been proposed and analyzed for their scaling behavior relative to classical sampling based motion planners [84], and a related line of work has examined quantum search approaches more broadly for sampling based motion planning [83], a topic we return to in Section 4 given its closer conceptual affinity to search than to optimization. Multi robot routing under real time constraints has similarly been explored using quantum computing technologies as an alternative to classical vehicle routing heuristics [29], and mobile robot order picking and batching in warehouse settings has been formulated and solved using a dedicated quantum computing approach [11]. Collision avoidance for UAVs has been addressed with a variational quantum algorithm [60], and improved quantum optimization algorithms have been applied to general path planning problems with reported gains over baseline quantum and classical heuristics [41]. Bilinear decomposition techniques have also been proposed to make mixed integer linear programs for multi agent motion planning tractable on near term quantum computers [3], and UAV tracking algorithms specifically have been evaluated for the performance gains offered by quantum computing based optimization [24]. A systematic review of quantum computing for automobile motion planning algorithms further situates

this literature within the broader intelligent transportation context [130], echoing a parallel and more general survey of quantum computing applications across transportation engineering [147].

Industrial and articulated robotic arm control forms a further cluster within quantum optimization, distinct from the mobile and aerial platforms discussed above in that the object being optimized is the arm's own joint space trajectory rather than its motion through an external environment. A quantum computing based algorithm for determining the orientation of robotic arms and rigid bodies has been proposed using quaternion representations [174], and this quaternion based approach has been extended to a full quantum model of forward kinematics built on quaternion to Pauli gate equivalence, applied to a six jointed industrial robotic arm [48]. Optimal control formulations combining quantum genetic algorithms with the same six jointed articulated arm have also been reported [31], alongside quantum inspired sliding mode control designed to improve both tracking precision and energy efficiency for articulated industrial arms [47]. Digital annealers, a restricted form of quantum inspired annealing hardware, have additionally been used to optimize the physical link length configuration between a human operator and a robot in leader follower collaborative operations [114]. Finally, quantum inspired hybrid artificial neural networks have been used to identify the dynamic parameters of mobile car-like robots [106], and quantum particle swarm optimization has been applied directly to proportional-derivative control tuning for trajectory tracking of the same class of car-like mobile robot [109], illustrating

that quantum inspired optimization is applied not only to a robot's path but to the low level controller gains that execute that path.

3.2. Task Allocation, Scheduling, and Kinematics

Task allocation and scheduling form a second major cluster. QAOA formulations with W-state initialization and XY-mixers have been used to decompose large multi drone assignment problems, translated from natural language instructions via a fine tuned language model, into hardware compatible sub-QUBOs [13]. Related QAOA and variational quantum eigensolver approaches have been proposed for multi robot task scheduling under real time adaptive control requirements [129], and quantum inspired meta-heuristics, including a quantum inspired elephant herd optimization algorithm, have been applied to coalition formation problems that arise in human robot collaborative assembly lines [99]. Quantum annealing has additionally been used to optimize robotic assembly line balancing directly, using the hybrid QBSolv algorithm to combine QUBO matrix partitioning with classical tabu search [159], and to optimize sum rate maximization in multi UAV aided wireless networks by jointly solving clustering, subchannel assignment, and power allocation as a combinatorial optimization problem [66]. At industrial scale, multi FPGA acceleration of simulated quantum annealing has been proposed to make large scale automated guided vehicle route optimization computationally tractable, addressing the qubit count limitations of physical annealers by simulating the annealing process on custom hardware [126].

Kinematics and posture optimization form a related cluster, distinct in that the QUBO formulation targets the robot's own configuration space rather than its trajectory through an external environment. Planar inverse kinematics for robotic manipulators has been reformulated as a QUBO using binary discretization of joint angles and solved on D-Wave hardware, with reported analysis of embedding choices across different annealer topologies [137]. A closely related QUBO formulation targets kinematic structure based robot design optimization itself, applied to a robotic hand case study where the combinatorial structure selection problem, rather than the motion of an already designed robot, is what gets solved on quantum hardware [70]. Robot posture optimization has also been demonstrated using quantum circuits for both forward and inverse kinematics on real quantum hardware, with the notable property that the qubit count scales with the number of robot links rather than with configuration space discretization, making a sixteen link humanoid representable with sixteen qubits [113].

3.3. Quantum-Inspired Metaheuristics and Synthesis

Beyond QUBO and QAOA proper, a substantial body of work uses quantum inspired classical metaheuristics, algorithms that borrow the mathematical language of superposition and quantum gates without requiring quantum hardware, to solve robot planning problems entirely on classical

computers. Quantum genetic algorithms have been applied to mobile robot path planning, using quantum bit encoding of routes and quantum crossover and mutation operators to improve convergence over classical genetic algorithms [98, 149]. A quantum inspired ant colony optimization algorithm has been used for robot coalition formation [172], and quantum behaved particle swarm optimization has been combined with fuzzy logic for adaptive parameter tuning in swarm robot navigation [95] and with grey wolf optimization for mobile robot obstacle avoidance path planning [165]. These quantum inspired approaches are worth distinguishing clearly from genuine quantum optimization, since they require no quantum hardware access at all and their reported gains come entirely from the heuristic structure of the search operator, not from any quantum computational advantage. The same quantum inspired pattern recurs across a wide range of further robotics control problems: quantum genetic algorithms have been used to evolve controllers for self reconfigurable modular robots [97] and to correct manipulator tracking error within a neural network based controller [81], quantum bacteria foraging algorithms have been applied to UAV parameter optimization and adaptive control [87], a hybrid quantum behaved particle swarm optimization and fuzzy logic approach has been proposed for dynamic path planning of nanoscale robots [102], and a quantum inspired sequence aware deep reinforcement learning algorithm has been proposed specifically for path planning [53], alongside a fuzzy A* variant that embeds quantum multi stage Q-learning within an artificial potential field planner [57]. Quantum inspired optimization has additionally been applied to adaptive path planning for cultural heritage site visits, a non-robotic but structurally identical routing problem [27], and comparative studies have directly benchmarked several quantum inspired control strategies, including sliding mode control, genetic algorithms, and Grover based search controllers, against one another on the same six degree of freedom manipulator [30]. Quantum inspired sliding mode control has also been proposed independently to improve energy efficiency and chattering suppression in DC motor driven robotic arms [77] and, in a related conference contribution, for high performance industrial robotic manipulation [46], while quantum fuzzy inference has been combined with cognitive intelligent control architectures for robotics and mechatronics more broadly [155]. Bipedal robot balance and obstacle avoidance has been addressed with a quantum computing based fuzzy PID controller [169], remote control of car-like mobile robots has been demonstrated via quantum teleportation circuits combined with quantum inspired PID tuning [107, 108], and a distinct remote robot control architecture has been proposed based on an alternative quantum computing model [116]. At the systems level, autonomous platforms more broadly have been proposed to draw on quantum inspired optimization algorithms as a route toward scalable onboard intelligence [20], quantum inspired methods have been applied to real time monitoring for autonomous vehicle and

drone Internet of Things systems [127], and a broader AI integrated quantum computing framework has been proposed for UAV system optimization [12]. Methane detection UAV path optimization has likewise been addressed with a hybrid quantum classical approach [124], and quantum enhanced AI robotics has been proposed for precision farming applications structurally similar to the outdoor navigation and resource scheduling problems addressed elsewhere in this section [74].

Taken together, this body of work indicates that quantum annealing has moved further toward practical robotics application than any other quantum computing paradigm discussed in this review, largely on the strength of its native fit to QUBO formulated planning and scheduling problems. However, nearly all reported results are either simulated, evaluated on small problem instances relative to real world deployment scale, or explicitly framed by their own authors as establishing feasibility baselines rather than demonstrating quantum advantage over well tuned classical solvers, a caveat we return to in Section 7.

4. Quantum Search and Sampling for Robotic Decision Making

Where Section 3 addressed problems with a well defined cost function to be minimized, this section turns to problems better described as search, finding a specific correct configuration within a large space, or sampling, estimating a distribution over possible states under uncertainty. Grover's algorithm and its variants dominate the search side of this literature, while quantum inspired sampling heuristics, largely classical in execution, dominate the estimation side. Table 3 summarizes representative studies from both branches.

4.1. Grover-Based Search for Kinematics and Localization

The clearest application of Grover's algorithm to robotics is manipulator kinematic optimization. Concretely, the oracle O_f referenced in Equation (4) and illustrated in Fig. 4(b) is constructed by marking candidate joint configurations x whose forward kinematics come within a tolerance ϵ of a target end-effector pose p^* ,

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle, \quad f(x) = \begin{cases} 1 & \text{if } \|FK(x) - p^*\| \leq \epsilon \\ 0 & \text{otherwise,} \end{cases} \quad (10)$$

where $FK(x)$ denotes the manipulator's forward kinematics map evaluated at joint configuration x . A parameterized quantum circuit is trained to approximate this forward kinematics model, and this trained circuit constructs the oracle O_f that Grover's algorithm then uses to identify optimal joint configurations, reporting speedups of up to ninety three times over classical Nelder-Mead optimization on simulated one and two degree of freedom manipulators as well as dual arm systems [105]. A related Grover operator based quantum variant of particle swarm optimization has been proposed

for the same inverse kinematics problem, using quantum amplitude amplification to identify a suboptimal particle that guides the swarm and thereby reducing the number of direct kinematics calculations required per iteration for manipulators with up to fifteen joints [138]. Grover's algorithm has also been applied outside the kinematics setting, most notably to robot localization, where it is used within adaptive Monte Carlo localization to perform search over horizontal and vertical position hypotheses in a discretized two dimensional map, validated both in simulation and on physical quantum computers [10]. A reinforcement learning based autonomous quantum agent built on a Grover based oracle has similarly been proposed and evaluated on a quantum game environment, demonstrating that amplitude amplification can reduce the number of observed states required to reach a solution relative to classical agents [7], and a Grover based approach to optimizing drone activation and coordination has combined a Grover oracle for marking energetically favorable configurations with a neural network trained on quantum sampled data [75].

Algorithm 2 summarizes the resulting pipeline in general form, applicable equally to the manipulator inverse kinematics setting of Nigatu and colleagues [105] and to the localization setting discussed above [10], differing only in how the oracle in Equation (10) is constructed and what quantity the decoded solution represents.

Algorithm 2 Generic Grover-Based Search Pipeline for Robotic Configuration Search

- 1: **Input:** search space of size N (discretized joint configurations or position hypotheses), target condition f , estimated number of marked solutions M
 - 2: Prepare an n -qubit register in uniform superposition, $|s\rangle = H^{\otimes n}|0\rangle^{\otimes n}$
 - 3: Construct oracle O_f marking states satisfying f (e.g. Equation (10) for manipulator IK)
 - 4: Construct diffusion operator $D = 2|s\rangle\langle s| - I$
 - 5: Compute iteration count $k^* \approx \frac{\pi}{4}\sqrt{N/M}$ following Equation (5)
 - 6: **for** $k = 1$ to k^* **do**
 - 7: Apply oracle: $|\psi\rangle \leftarrow O_f|\psi\rangle$
 - 8: Apply diffuser: $|\psi\rangle \leftarrow D|\psi\rangle$
 - 9: **end for**
 - 10: Measure the register to obtain candidate configuration x^* with high probability
 - 11: Decode x^* into the corresponding manipulator joint configuration or robot position estimate
 - 12: **Output:** decoded configuration x^* and number of oracle queries used
-

4.2. Search in Multi-Agent Settings

Search structure also appears in multi agent settings. A quantum multi agent reinforcement learning framework for swarm robot navigation uses the Grover algorithm for action decision making within a cooperative navigation task, alongside a quantum based prioritized experience replay method

Table 3

Representative quantum search and quantum inspired sampling studies for robotic decision making.

Problem Type	Year	Study	Method Type	Key Idea
Manipulator IK	2025	[105]	Grover + QML oracle	Up to 93x speedup over classical Nelder-Mead in simulation
Manipulator IK	2025	[138]	Grover operator + PSO	Amplitude amplification guides swarm for 7-15 joint manipulators
Robot navigation	2022	[85]	Quantum maze solving algorithm	Grover-type search applied directly to maze navigation
Sampling-based planning	2023	[83]	Quantum search framework	Extends quantum RRT into a search theoretic framing
Robot motion planning	2022	[22]	Quantum planner	Planning problem formulated to exploit quantum search structure directly
Localization	2025	[10]	Grover's algorithm	Search over position hypotheses within AMCL
Localization (heuristic)	2026	[5]	Quantum-inspired localization (heuristic)	Indoor localization, no quantum hardware
Vehicle self-localization (heuristic)	2022	[168]	Quantum-inspired evolutionary algorithm (heuristic)	Dead reckoning drift calibration, no quantum hardware
Soccer robot localization (heuristic)	2014	[56]	Quantum immune algorithm (heuristic)	Real-time self-localization, no quantum hardware
Autonomous agent	2025	[7]	Grover based RL agent	Fewer observed states to reach solution than classical agents
Drone activation	2026	[75]	Grover oracle + neural network	Quantum sampled training data for classical prediction
Multi agent navigation	2024	[26]	Grover + prioritized replay	Action decision making in cooperative navigation
SLAM	2018	[175]	Quantum-behaved PSO (heuristic)	Improves particle filter proposal distribution, no quantum hardware
Distributed SLAM	2016	[120]	Quantum-behaved PSO (heuristic)	Replaces local filters in distributed SLAM, no quantum hardware
Swarm navigation	2025	[95]	Quantum-behaved PSO + fuzzy logic (heuristic)	Adaptive parameter tuning, no quantum hardware

intended to address the degraded performance that classical experience replay techniques exhibit as the number of agents grows [26]. This reflects a broader pattern in the literature in which Grover type search is most often embedded as one component of a larger hybrid pipeline, providing the discrete search step, rather than serving as a standalone algorithm applied directly to a raw robotics problem.

A related and conceptually important line of work treats robot navigation itself as a maze solving or graph search problem to which Grover type quantum search can be applied directly, rather than embedding search as a subroutine within a larger optimization pipeline. A dedicated quantum maze solving algorithm has been developed and validated on a small scale maze navigation task [85], and quantum search approaches more broadly have been analyzed for their applicability to sampling based motion planning, extending parallel implementations of rapidly exploring random trees discussed in Section 3 into a search theoretic framing [83]. A dedicated quantum planner for robot motion has similarly been proposed, formulating the planning problem in a way intended to exploit quantum search structure directly rather than through a classical planner augmented with a quantum subroutine [22].

Localization itself has attracted a second, largely quantum inspired thread distinct from the genuinely quantum Grover based localization method discussed above. Quantum inspired robot localization methods have been proposed for indoor environments using classically executed heuristics that borrow quantum probability formalism [5], and a dead reckoning calibration scheme based on an adaptive quantum inspired evolutionary algorithm has been developed for vehicle self localization, correcting drift accumulated through inertial dead reckoning using a classically executed quantum inspired optimizer [168]. Quantum immune algorithms, a further quantum inspired metaheuristic family, have been applied to the self localization problem for omni-directional soccer robots operating under the tight real time constraints of robot soccer competition [56]. As with the SLAM literature discussed below, these localization methods illustrate that the term quantum in robot localization research spans a spectrum from genuine Grover search on physical quantum hardware to purely classical heuristics inspired by quantum probability, and the distinction matters for assessing what has actually been demonstrated.

4.3. Quantum-Inspired Sampling for SLAM

Sampling problems in robotics are dominated by simultaneous localization and mapping, and here the literature diverges sharply from the search literature in one important respect: almost none of the reported work executes on quantum hardware. Instead, the dominant approach borrows the mathematical formalism of quantum mechanics as a classical optimization heuristic. Quantum behaved particle swarm optimization has been used to improve the proposal distribution in Rao-Blackwellized particle filter based SLAM, directly addressing the particle degeneracy and impoverishment problems that arise when a small number of high weight particles are repeatedly duplicated during resampling [175]. The same quantum behaved particle swarm optimization heuristic has been extended to distributed SLAM systems, where it replaces the local filters used at each node in a distributed architecture for mobile robot navigation [120]. Quantum behaved particle swarm optimization has also been combined with fuzzy logic based adaptive parameter tuning for swarm robot navigation more broadly, adjusting convergence behavior in response to context based evaluation metrics rather than fixed parameters [95].

This gap between search and sampling is worth stating plainly, since it is easy for the shared vocabulary of quantum optimization to obscure it. Grover based search methods for robotics, while still confined to small problem instances and simulation, do in a meaningful number of cases execute on physical quantum hardware and report results specific to genuine quantum circuits. Quantum behaved sampling methods for SLAM, by contrast, are almost entirely quantum inspired classical algorithms whose connection to quantum computation is terminological rather than computational, reformulating a classical particle swarm optimizer using the language of quantum superposition without requiring, or benefiting from, any quantum hardware. Both lines of work are legitimate contributions to their respective problems, but conflating them risks overstating how far genuinely quantum methods have progressed toward solving robotic state estimation under uncertainty, which remains, at the time of writing, largely unaddressed by quantum computation proper.

5. Quantum Machine Learning in Robotics

Quantum machine learning applies parameterized quantum circuits, trained in a manner analogous to classical neural networks, to perception, policy learning, and control. In robotics, the largest and most mature application of this approach is quantum reinforcement learning, with a lineage stretching back nearly two decades before the recent wave of variational quantum circuit based work. Table 4 traces this lineage from early quantum inspired heuristics through recent variational and perception focused studies.

5.1. Quantum Reinforcement Learning: From Quantum-Inspired Heuristics to Variational Circuits

The earliest thread in this lineage is quantum inspired reinforcement learning, in which the reinforcement learning update rule itself is reformulated using quantum probability amplitudes without requiring quantum hardware. A quantum robot architecture combining quantum structure with reinforcement learning algorithms was proposed early in this line of work [38], followed by quantum reinforcement learning formulations using amplitude amplification to bias action selection toward previously successful actions [32, 35], and incoherent control methods for quantum systems with controllable subspaces trained via quantum reinforcement learning [36]. This line matured into a robust quantum inspired reinforcement learning algorithm for robot navigation, in which a probabilistic action selection policy inspired by the quantum measurement collapse phenomenon was shown to be more robust to learning rate and initial state choices than classical reinforcement learning, both in simulation and on a physical mobile robot [37]. A related quantum parallelization of hierarchical Q-learning has been proposed for global mobile robot navigation, using quantum parallelization to bridge local reactive control and topological navigation levels within a single accelerated learning process [23].

The more recent thread uses genuine variational quantum circuits, parameterized quantum gates trained via classical gradient based optimization, as function approximators within a reinforcement learning loop. Quantum deep reinforcement learning has been used to train navigation policies for a wheeled robot across simulated environments of increasing complexity, comparing consecutive versus per qubit encoding strategies for the classical state vector and reporting that the quantum circuits required substantially fewer trainable parameters than a classical neural network baseline while remaining less stable in training and slightly less robust when tested in larger dynamic environments [54]. A variational quantum soft actor critic architecture has been proposed for continuous robotic arm control, replacing the actor network of the classical soft actor critic algorithm with a variational quantum circuit and evaluating the approach via digital quantum circuit simulation against a classical benchmark [2]. A related quantum enhanced reinforcement learning framework quantizes both the states and actions of the learning agent and introduces a probability amplification method intended to address the exploration exploitation trade off directly [58]. These variational approaches draw on a broader body of quantum reinforcement learning research developed outside robotics specifically, including quantum enhancements for deep reinforcement learning in large state spaces [67], variational quantum circuit designs benchmarked on stochastic environments such as frozen lake [40], and general purpose variational quantum circuits for deep reinforcement learning [25], all of which have informed the design choices made in the robotics specific work surveyed here [96].

Table 4

Representative quantum machine learning studies in robotics, spanning quantum inspired reinforcement learning, variational quantum circuits, and perception applications.

Category	Year	Study	Hardware Basis	Key Idea
QRL foundations	2006	[38]	Quantum inspired (heuristic)	Early quantum robot architecture combining structure with RL
QRL foundations	2008	[35]	Quantum inspired (heuristic)	Amplitude amplification biases action selection
QRL foundations	2012	[37]	Quantum inspired (heuristic)	Robust navigation policy on simulation and physical robot
QRL foundations	2012	[23]	Quantum inspired (heuristic)	Quantum parallelization of hierarchical Q-learning
Variational QRL	2024	[54]	Variational quantum circuit	Fewer trainable parameters than classical baseline for navigation
Variational QRL	2022	[2]	Variational quantum circuit	Quantum soft actor-critic for continuous arm control
Variational QRL	2021	[58]	Quantum inspired (heuristic)	Quantized states/actions with probability amplification
Multi-agent QRL	2024	[150]	Variational quantum circuit	Hybrid QVC + spiking network for MARL in autonomous robotics
Perception	2024	[44]	Quantum ANN (simulated)	Singularity avoidance in manipulator IK
Perception	2021	[16]	Quaternion quantum NN	Perception and control of robot manipulators
Perception	2025	[158]	Quantum reservoir computing	Test oracle for autonomous mobile robot regression testing
Perception	2025	[73]	Quantum enhanced transformer	Energy transfer modeling in physical human-robot interaction
Perception	2024	[134]	Quantum CNN	Image classification from UAV footage

Training the variational circuits underlying this thread requires estimating the gradient of Equation (7) with respect to the circuit parameters θ , since standard backpropagation through a physical quantum circuit is not available. The parameter-shift rule provides an analytic, hardware-compatible gradient estimator for circuits built from single-qubit rotation gates,

$$\frac{\partial \langle O \rangle_{\theta}}{\partial \theta_i} = \frac{1}{2} \left[\langle O \rangle_{\theta_i + \pi/2} - \langle O \rangle_{\theta_i - \pi/2} \right], \quad (11)$$

which evaluates the same circuit twice, with the i -th parameter shifted by $\pm\pi/2$, and requires no access to the circuit's internal quantum state. This gradient is then combined with the policy gradient objective from classical reinforcement learning, typically the REINFORCE estimator

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R_t \right], \quad (12)$$

where τ denotes a trajectory of states and actions and R_t is the return from time step t onward, with π_{θ} given by the quantum circuit output of Equation (7) and ∇_{θ} computed via Equation (11). Algorithm 3 summarizes the resulting hybrid training loop shared, with minor variation, across the variational quantum reinforcement learning studies in Table 4.

5.2. Multi-Agent Extensions

Multi agent extensions of this work address coordination among several robots or agents simultaneously. A hybrid

framework combining quantum variational circuits for high level policy exploration with spiking neural networks for low latency motor control has been proposed for safe and explainable multi agent reinforcement learning in autonomous robotics [150], motivated explicitly by the growing complexity of coordinating self driving vehicles, unmanned aerial vehicles, and collaborative robotic swarms.

The variational quantum reinforcement learning literature surveyed above sits within a broader and still rapidly evolving body of quantum reinforcement learning research, only part of which has yet been applied specifically to robotic control. Continuous quantum reinforcement learning formulations, in contrast to the discretized action spaces common in earlier work, have been proposed specifically for robot navigation tasks [39], and quantum reinforcement learning has also been applied to vehicle obstacle avoidance and motion planning, framing collision avoidance as a policy learning problem solved with a quantum augmented agent [171]. Outside robotics proper, quantum reinforcement learning has been benchmarked on board games [153], and broader reviews have traced the progress of reinforcement learning methods as they incorporate quantum computing techniques [71], complementing the robotics specific quantum reinforcement learning survey material summarized in Table 4. A broader survey of hybrid quantum-classical machine learning research in the NISQ era situates these robotics specific reinforcement learning studies within the wider landscape of near term quantum machine learning [33].

Algorithm 3 Generic Variational Quantum Reinforcement Learning Training Loop

```

1: Input: robot environment with state space  $S$ , action
   space  $A$ ; parameterized quantum circuit  $U(\theta)$ ; classical
   state encoding map  $\phi$ 
2: Initialize circuit parameters  $\theta$  randomly
3: for each training episode do
4:   Observe classical robot state  $s_t$  (e.g. pose, sensor
   readings)
5:   Encode  $s_t$  into a quantum state  $|\phi(s_t)\rangle$  (e.g. angle or
   amplitude encoding)
6:   Evaluate the circuit to obtain  $\pi_\theta(a | s_t)$  via Equa-
   tion (7)
7:   Sample action  $a_t \sim \pi_\theta(\cdot | s_t)$  and execute on the
   robot or in simulation
8:   Observe reward  $r_t$  and next state  $s_{t+1}$ ; store transi-
   tion
9:   if episode complete then
10:     Compute returns  $R_t$  for all  $t$  in the episode
11:     Estimate  $\nabla_\theta J(\theta)$  via Equation (12), using the
     parameter-shift rule (Equation (11)) for each component
12:     Update  $\theta \leftarrow \theta + \eta \nabla_\theta J(\theta)$  via a classical
     optimizer with learning rate  $\eta$ 
13:   end if
14: end for
15: Output: trained circuit parameters  $\theta$  defining the robot
   control policy

```

5.3. Quantum Machine Learning for Perception and Estimation

Beyond reinforcement learning, quantum machine learning has been applied to perception and estimation tasks more directly. Quantum artificial neural networks have been investigated for singularity avoidance in robotic manipulator inverse kinematics, reporting reductions in mean absolute error relative to classical artificial neural networks for both position and orientation prediction [44]. Quaternion valued quantum neural networks have been proposed for perception and control of robot manipulators, exploiting the natural fit between quaternions and three dimensional rotation to extend real valued spiking neural network architectures [16]. Quantum machine learning has also been used to construct a test oracle for regression testing of autonomous mobile robots, combining quantum reservoir computing with a classical neural network to predict expected robot behavior and reporting a measurable reduction in prediction error relative to a purely classical baseline [158]. Quantum enhanced transformer networks have been proposed to model energy transfer dynamics during physical human robot interaction in gait rehabilitation robotics [73], and a related passivity based neuromechanical control framework combines a passivity observer with quantum enhanced deep reinforcement learning for adaptive impedance scaling in wrist rehabilitation [72]. Quantum machine learning has further been applied to image classification from unmanned aerial vehicle footage [134], to quantum inspired multi scale object detection for

ultra small objects in UAV imagery [104], to layer by layer learning for image processing in UAV applications [145], to tactical augmented reality systems that rely on multi robot sensor fusion in military contexts [110], to adaptive stereo vision for autonomous robotics [151], to autonomous firefighting robots combining fire detection with quantum enhanced decision making [148], to telesurgery training systems that combine quantum machine learning with extended reality [55], and to exoskeleton and aquatic rehabilitation therapy systems that integrate quantum computing into the control loop [135]. A quantum computing and artificial neural network hybrid has similarly been proposed for general classification and time series prediction tasks with agricultural robotics relevance [4], and quantum machine learning has been proposed for space applications sharing the same autonomy and perception demands as terrestrial field robotics [103]. Quantum machine learning has also begun to be examined specifically for feasibility on embedded and onboard robotic hardware, with recent work assessing hybrid architectures and dedicated quantum co-processors for embedded systems deployment [34], and optoelectronic robotic systems have been proposed as a target application domain for precision focused quantum machine learning [161].

A smaller but conceptually distinct thread applies quantum formalism not to perception or control directly but to modeling a robot's internal affective or cognitive state. Quantum structures have been proposed for modeling the emotion space of robots [163], extended into a broader conceptual framework for quantum affective computing intended to fuse emotional signals across multiple robots [162]. Related work has proposed quantum modeling approaches for reactive agents more generally [78], minimal variable quantum decision makers for robotic control [79], and has explored, at a more speculative level, whether quantum formalism can illuminate robotic creativity and survival oriented behavior [91] or support a dialogue between classical and quantum accounts of robotic thinking and creativity [89]. Closely related experimental work has proposed a classical mechanical apparatus that reproduces certain quantum computational behaviors, including a method for measuring the weighting factor of eigenstates in quantum superposition without genuine quantum hardware, and has speculated on the relevance of such classical quantum analogues to artificial intelligence mounted on robots [64, 65]. These contributions sit at the boundary of what this review considers within scope, since their connection to measurable robotic task performance is less direct than the perception and control literature discussed above, but they are included because they illustrate a distinct, cognitively motivated rationale for applying quantum computing to robotics, oriented toward representing internal robot states rather than solving external computational problems, echoed by an early proposal for modeling emotional robot behavior directly with quantum mechanical formalism [88], by a quantum inspired bidirectional associative memory architecture for human robot interaction [94], by sonification work that renders the

trajectories of a quantum driven robotic swarm as an auditory feedback signal [92], by a related case study applying category theory alongside quantum computing to swarm robotics [90], and, at the most exploratory end of this thread, by an early outreach effort introducing quantum robotics concepts to teenage students through simple physical robot demonstrations [128].

Foundational quantum machine learning results developed outside the robotics literature underpin much of this applied work. Early experimental demonstrations of entanglement based classification on small scale photonic quantum computers established the working principle of quantum enhanced supervised and unsupervised learning [19], and subsequent surveys have mapped both the near term outlook for supervised quantum machine learning on noisy intermediate scale hardware [154] and the broader landscape of quantum enhanced machine learning applications [140]. As with quantum optimization, the reinforcement learning results surveyed in this section report parameter efficiency gains, meaning fewer trainable weights for comparable task performance, considerably more often and more convincingly than they report outright gains in training speed, final task performance, or robustness relative to well tuned classical baselines, a pattern consistent with the broader NISQ era caveat discussed further in Section 7.

6. Hardware, Latency, and Systems Constraints

6.1. Onboard Computing and Latency

The results surveyed in Sections 3 through 5 must be read against the hardware they were obtained on, since almost every reported gain is qualified, explicitly or implicitly, by the constraints of noisy intermediate scale quantum devices. A structured assessment of what current quantum hardware can plausibly offer onboard robotic computer systems concludes that immediate onboard deployment of quantum coprocessors is unrealistic given current size, weight, power, and reliability constraints, and instead proposes a staged path beginning with near term offboard or cloud assisted quantum processing for pre and post processing steps, progressing to medium term hybrid classical quantum pipelines, and only in the longer term integrating compact quantum coprocessors directly onto robotic platforms [18]. This staged framing is consistent with the pattern observed throughout this review, in which nearly all quantum annealing and gate based results are obtained either in simulation or via cloud access to remote quantum hardware, with the classical control loop and real time execution remaining entirely classical.

Latency is the specific constraint that most directly limits robotics applications relative to other quantum computing application domains. Robotic control loops typically require decisions on the order of milliseconds, while quantum annealing access times, gate based circuit execution, and especially the classical overhead of embedding a problem onto

physical qubit topology, can introduce delays that are incompatible with real time control even when the underlying optimization itself is fast. Multiple object tracking has been used as a case study for this constraint directly, using reverse annealing to reduce the time to solution of quantum annealing based tracking by over ninety nine percent relative to conventional quantum annealing implementations, achieving annealing times on the order of microseconds per tracking process and explicitly framing real time applicability, including collision prediction for autonomous robots and vehicles, as the motivating goal [62]. Multi FPGA acceleration of simulated quantum annealing represents a complementary approach to the same latency problem, sidestepping physical qubit count limits entirely by simulating the annealing process on custom digital hardware, and has been demonstrated on large scale automated guided vehicle routing problems with tens of thousands of variables [126].

6.2. Software, Systems, and Security Infrastructure

Software and systems level infrastructure for quantum robotics remains comparatively immature. Frameworks intended to bridge classical and quantum software development for robotics style applications have begun to appear, addressing the practical problem that most robotics engineers have no experience writing quantum circuits and most quantum computing tools have no awareness of robotics specific data structures or real time constraints [166]. Security of the communication layer in quantum augmented robotic systems has also been addressed directly, with a proposed architecture using quantum cryptography based on the BB84 protocol to secure network communication within the Robot Operating System 2 ecosystem, motivated by the observation that data messages in that ecosystem are transmitted in plain text by default [119]. Earlier work on quantum software engineering for intelligent robotics addressed the more foundational problem of efficient circuit implementation for the quantum algorithmic gates needed to support massively parallel quantum computation on classical simulators, an important precursor to running any of the algorithms surveyed in this review at meaningful scale [76].

6.3. Quantum Sensing as a Complementary Technology

Quantum sensing is a closely related but conceptually distinct technology from the quantum computing methods discussed elsewhere in this review, and it is included here rather than as a standalone section because its role in robotics is primarily as a hardware layer that feeds classical or quantum enhanced estimation pipelines, rather than as a computational method in its own right. Quantum sensors exploit superposition and entanglement to achieve absolute, drift free measurements of inertial forces, gravity, and magnetic fields, and a comprehensive review of the technology identifies cold atom interferometry, atomic vapor cells, superconducting quantum interference devices, and nitrogen vacancy

Table 5

Quantum sensing modalities relevant to robotic navigation and positioning, summarized from the sources discussed in this section.

Modality	Physical Principle	Reported Capability	Primary Deployment Barrier
Cold atom interferometry	Light pulse atom interferometry	Navigation grade position estimation without external reference [15, 50]	Miniaturization and ruggedization for mobile platforms [152]
Hybrid quantum-classical accelerometry	Atom interferometry + classical Kalman filtering	Practical navigation-compatible inertial sensing [21]	Integration overhead with classical filtering pipelines
Atomic vapor cells / SQUIDs / NV centers	Superposition and entanglement based field sensing	Sub-meter accuracy, drift-free inertial/magnetic measurement [86]	Laboratory-scale equipment historically required
Classical-quantum sensor fusion	Combined classical inertial sensors with quantum sensor drift correction	Wider bandwidth and range than quantum sensors alone [4]	Systems integration and cost

centers in diamond as the leading modalities for positioning and navigation applications, with reported ability to reduce navigation drift and achieve sub meter accuracy in both standalone and hybrid quantum classical architectures [86]. Table 5 summarizes these modalities alongside their reported capabilities and primary deployment barriers.

This capability is particularly relevant for robotics operating in GPS denied or GPS contested environments, since quantum inertial navigation systems built on light pulse atom interferometry can in principle provide navigation grade position estimates without external reference signals [15, 50], and hybrid quantum classical accelerometers combining atom interferometric measurements with classical Kalman filtering have been demonstrated as a practical path toward navigation compatible quantum inertial sensing [21]. Industry and government laboratory sources report that the primary barriers to deploying quantum sensors on dynamic robotic platforms are miniaturization and ruggedization rather than fundamental sensing precision, since isolating and controlling atoms for interferometry has historically required laboratory scale equipment incompatible with mobile platforms [152], motivating current work on classical quantum sensor fusion architectures that combine the bandwidth and range of classical inertial sensors with the drift free accuracy of quantum sensors.

6.4. Adjacent Domains: Autonomous Vehicles and Energy Efficiency

Adjacent application domains illustrate both the promise and the immaturity of quantum computing for time sensitive, safety critical autonomous systems more broadly. Reviews of quantum computing for advanced driver assistance systems and autonomous vehicles identify latency, computational speed, algorithmic accuracy, security, and privacy as the primary limitations of classical approaches that quantum machine learning is proposed to address, while acknowledging that most reported results remain at the proof of concept stage [133], a conclusion echoed by Qiskit based

frameworks combining quantum neural networks for trajectory prediction with quantum approximate optimization for vehicle routing [6]. Energy efficiency is a further systems level constraint that has begun to receive direct attention, including energy minimization frameworks for quantum federated learning over low altitude UAV networks that jointly optimize quantum computing power, transmit power, and trajectory [142], and broader proposals to integrate quantum computing with generative artificial intelligence for energy efficient cyber physical systems spanning robotics, smart grids, and autonomous transportation [141]. Security specific to quantum threats has also emerged as a systems concern in its own right, with post quantum authentication protocols proposed for UAV assisted vehicular networks that treat UAVs explicitly as untrusted relay nodes under the assumption that classical public key cryptography will eventually be broken by large scale quantum computers [118], and blockchain assisted quantum optimization frameworks proposed for collision free UAV coordination in zero trust airspace environments [132].

UAV networking and swarm coordination represent a further hardware adjacent constraint, since many of the reported quantum benefits for aerial robotics are bound up with the wireless communication infrastructure connecting individual platforms rather than with the onboard computation of any single vehicle. A broader review situates the internet of quantum drones as an emerging networking paradigm, cataloging the security, coordination, and communication challenges that arise once quantum resources are distributed across a fleet of aerial robots rather than concentrated on a single processor [80]. More recent work has proposed quantum machine learning specifically for wireless powered UAV positioning within sixth generation digital twin architectures supported by cooperative nano-satellite constellations [123], and quantum-classical adaptive navigation frameworks combining QAOA with proximal policy optimization have been proposed for UAV navigation in GNSS denied urban environments, addressing a scenario where

classical satellite based positioning is unavailable and where quantum sensing, discussed above, offers a complementary but separate solution path [144]. These networking and positioning constraints compound, rather than substitute for, the onboard latency and qubit count constraints discussed earlier in this section, since a quantum optimization result computed with acceptable latency onboard a single robot may still be delayed or degraded by the wireless link connecting that robot to the rest of its swarm or to a cloud based quantum resource.

Robot evaluation and human robot interaction have also begun to draw on quantum formalism as a decision support tool rather than as a computational accelerant. Quantum probability combined with fuzzy logic has been proposed as a multi criteria decision making framework specifically for evaluating candidate robots against multiple competing criteria [115], illustrating that quantum inspired methods in this literature are applied not only to a robot's own control and perception but to the human decision processes of selecting, configuring, and evaluating robotic systems. Security of robotic systems against classical and emerging cyber threats has also drawn on quantum and quantum inspired methods, including artificial intelligence and machine learning based classification of robotic and cyberattack traffic [1], a quantum cellular neural network combined with chaotic map encryption for securing robotic image data [157], and quantum computation combined with machine learning for smart intelligent control applications in adversarial network settings [146]. Finally, at a more exploratory hardware boundary, early work on brain quantum-computer interfaces has speculated about eventual applications to robot control mediated directly by a human operator's neural signals [69], illustrating a hardware interface possibility that remains well outside current deployment but is occasionally discussed alongside the other systems level constraints surveyed in this section.

7. Open Challenges and Future Directions

Several challenges recur across the optimization, search, sampling, and learning literatures surveyed in this review, and addressing them will largely determine whether quantum computing in robotics matures from a proof of concept research area into a deployed technology. Fig. 7 summarizes these challenges alongside the future directions discussed at the end of this section.

7.1. Reliance on Simulation over Physical Validation

The most consistent limitation is the near total reliance on simulation rather than physical validation, or, where physical quantum hardware is used, validation on small problem instances far below the scale required for real world robotic deployment. This pattern holds across quantum annealing, gate based, and quantum machine learning approaches alike, and it means that reported speedups, whether the ninety three times reported for Grover assisted manipulator optimization [105] or the parameter efficiency gains reported for variational quantum reinforcement learning [54],

should be read as evidence of feasibility and problem structure fit rather than as demonstrated advantage over well engineered classical solvers at deployment scale. Several of the papers surveyed here are explicit about this limitation themselves, framing their contribution as establishing a baseline or characterizing failure modes rather than claiming superiority over classical methods [137, 156].

7.2. Absence of Standardized Benchmarks

A second challenge is the absence of standardized, application level benchmarks specific to robotics. General purpose quantum computing benchmarking frameworks have begun to address this gap, including one that explicitly formulates robot path optimization as a reference problem alongside vehicle option optimization from the industrial domain [49], and broader surveys of quantum computing for scheduling problems have called for grounding quantum algorithm development in empirical case studies rather than purely theoretical complexity analysis [8]. Without shared benchmarks, it remains difficult to compare results across the fragmented literature this review has organized by computational primitive, since papers differ simultaneously in problem instance size, hardware platform, classical baseline choice, and evaluation metric.

7.3. Conceptual Ambiguity of the Term Quantum

A third challenge concerns the conceptual clarity of the term quantum itself within this literature. As discussed in Sections 2 and 4, a substantial share of published work on quantum robotics uses quantum formalism as a source of classical algorithmic heuristics, quantum behaved particle swarm optimization and quantum genetic algorithms being the clearest examples, without any quantum hardware execution at all. This work has genuine value as a heuristic optimization technique, but its inclusion alongside genuinely quantum hardware results under the same umbrella term risks overstating the field's actual progress toward quantum advantage. Future surveys and, more importantly, future primary research would benefit from consistently distinguishing quantum inspired classical methods from quantum executed methods, a distinction this review has attempted to maintain throughout.

7.4. Real-Time and Latency Constraints

A fourth and closely related challenge is the mismatch between the real time demands of robotic control and the latency profile of current quantum hardware, discussed at length in Section 6. Reverse annealing and hardware acceleration approaches have begun to narrow this gap for specific tasks such as object tracking [62, 126], but a general solution for embedding quantum computation within a hard real time robotic control loop remains an open problem, and the staged deployment path proposed by Bocharov, moving from offboard cloud assistance toward eventual onboard integration [18], reflects a field wide consensus that this constraint will not be resolved by algorithmic improvements alone.

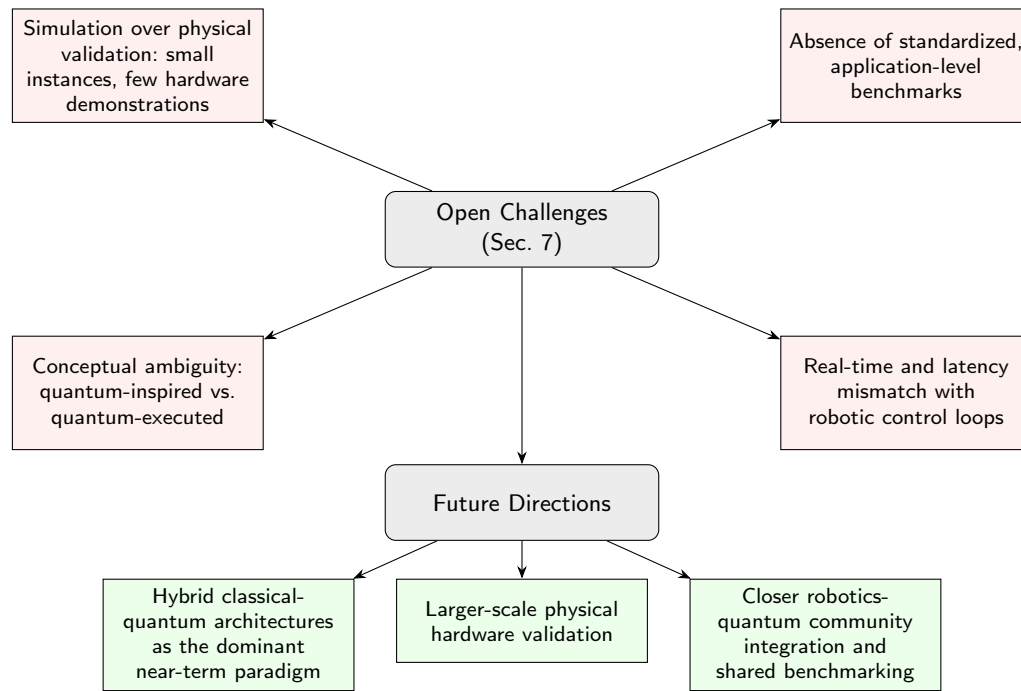


Figure 7: Overview of the open challenges identified in this review and the future directions they motivate.

7.5. Future Directions

Looking forward, several directions appear most likely to yield near term progress. Hybrid quantum classical architectures, in which quantum hardware handles a bounded subroutine embedded within an otherwise classical pipeline, already dominate the literature surveyed in this review and are likely to remain the dominant paradigm as quantum hardware scales gradually rather than discontinuously. Larger scale physical hardware validation, moving beyond the small manipulator and grid world instances common in current work toward problem sizes representative of actual deployment, would substantially strengthen the evidentiary basis for claims of quantum benefit. Finally, closer integration between the robotics and quantum computing research communities, evidenced by recent introductory surveys aimed explicitly at bridging the two fields [28, 45, 164], appears necessary to ensure that future quantum algorithm design is informed by the actual computational bottlenecks robotics practitioners face, rather than by problem formulations chosen primarily for their fit to available quantum hardware.

8. Conclusion

This review has organized the literature on quantum computing in robotics around the computational role each robotics problem plays, namely optimization, search, sampling, and learning, rather than around quantum algorithm families alone. Quantum annealing has progressed furthest toward practical application, owing to its native fit with QUBO formulated planning, scheduling, and kinematics problems. Grover based search methods have demonstrated meaningful, if small scale, results on physical quantum

hardware for kinematic optimization and localization. Quantum machine learning, particularly quantum reinforcement learning, has produced a substantial and maturing body of work, most convincingly demonstrating parameter efficiency rather than outright performance gains over classical baselines. Sampling based estimation problems such as SLAM remain the least addressed by genuinely quantum methods, with the existing literature dominated instead by quantum inspired classical heuristics. Across all four categories, the binding constraint is not primarily algorithmic but architectural: current noisy intermediate scale quantum hardware, whether annealer or gate based, cannot yet meet the size, latency, and reliability requirements of deployed robotic systems, and the field's own most careful papers acknowledge this directly. The path forward, as reflected throughout the literature surveyed here, runs through hybrid classical quantum architectures, larger scale physical validation, and standardized benchmarking, rather than through any single quantum algorithmic breakthrough. As both quantum hardware and the robotics specific software tooling needed to interface with it continue to mature, the computational taxonomy offered in this review should remain a useful lens for assessing which robotics problems are genuinely well suited to quantum approaches, and which are not.

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